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Fast Optimal Sizing of Domestic PV Energy Systems*

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Abstract

The increasing adoption of distributed Solar Photovoltaic (PV) systems presents a significant opportunity to decarbonize residential buildings and enhance energy resilience. However, determining the optimal size of a PV system, when coupled with battery storage, remains a complex challenge influenced by various factors including energy consumption, solar irradiance and the desired level of self-sufficiency. While current approaches rely on detailed simulations to estimate optimal system sizes, these methods are computationally intensive and may require several hours to complete. Instead, we propose a neural network-based approach as a fast and accurate alternative, capable of generating sizing recommendations within milliseconds, approximately 40,000 times faster than the baseline. By significantly reducing computation time, this approach aims to support consumers, planners, and policymakers in maximizing the economic and environmental benefits of PV deployment.

CCS Concepts

• Hardware → Power and energy.

Keywords

Energy Storage, Solar Power, Sizing, Neural Networks

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1 Introduction

The urgency of addressing climate change and the increasing costs associated with traditional energy sources are driving a global shift towards renewable energy technologies, particularly solar PV [7]. Distributed solar PV systems, installed at residential, commercial, or community levels, offer a pathway to cleaner energy generation, reduced electricity bills, and enhanced energy independence. However, the intermittent nature of solar energy requires careful consideration of system sizing to meet energy demands reliably.

Oversizing a PV system can result in underutilized generation and unnecessarily high upfront costs, while undersizing may lead

to insufficient self-consumption and continued reliance on grid electricity which may originate from non-renewable sources. The inclusion of battery storage adds further complexity, requiring a joint optimization of PV and storage capacities [8]. Importantly, the overall cost of the system should be minimized to encourage broader adoption of renewable energy solutions.

2 Problem Statement

Conventional methods for determining optimal system sizes rely on detailed simulations that, while accurate, can be time-consuming and computationally expensive [5]. To address this, we explore a data-driven approach using a multilayer perceptron (MLP) neural network to predict the optimal PV and battery sizes based on user-specific load profiles and solar generation potentials. This approach aims to deliver near-instant sizing recommendations, ultimately supporting widespread adoption of distributed solar PV systems.

3 Methodology

We propose using an MLP to predict the optimal sizing of PV and battery systems for residential buildings. The model is trained on domestic electricity load profiles from the UK and solar generation estimates from PVWatts [9], both spanning hourly resolution over a full year. Ground-truth labels for optimal PV and battery sizes are generated using a single-roof simulator [4, 8].

A total of 1000 initial load profiles were generated using *Faraday*, a generative AI model capable of producing synthetic smart meter data conditioned on specific household characteristics [1]. The dataset reflects a diverse range of household archetypes within the UK, including EV ownership in 20% of the samples where EVs are assumed to be charged primarily at night. To augment the dataset, 2,000 additional load profiles were generated by applying a random shift between -0.1 kWh and 0.3 kWh to the original profiles, simulating variations in electricity consumption. Gaussian noise, with a mean of 0.0 kWh and a standard deviation of 0.1 kWh, was then added to introduce realistic fluctuations. To prevent abrupt or unnatural peaks, the noise at each time step was smoothed by averaging it with the noise from the preceding time step.

Solar generation profiles were obtained using *PVWatts* [9] which computes the estimated production over a year for a given location based on historical weather data. In total, 419 solar profiles from various geographic locations around the world were collected. To increase dataset diversity, especially under varying weather conditions, we augment the data by duplicating solar profiles by bridging and creating gaps in solar curves. This augmentation leverages the fact that cloud coverage is highly variable over the years which may result in such gaps in solar curves. Additionally, the solar profiles were scaled by a random factor between -20% and 20% to model higher or lower generation as seen in Figure 1.

* Code available at https://github.com/juliagsch/PV_Battery_Sizing.

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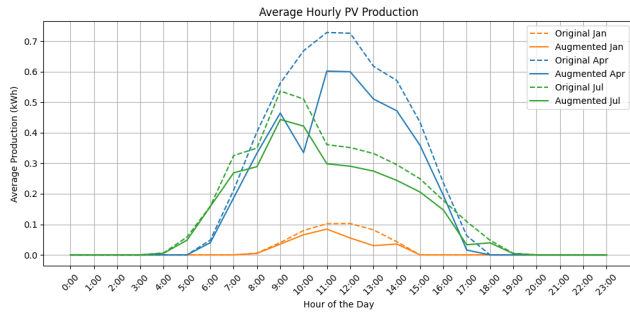


Figure 1: PV production curves for a 1 kW system in London, UK for random days in January, April and July.

The data is split into a train and test set using a 90/10 split. Then, the final dataset is constructed by randomly pairing each load profile with six different solar profiles, generating 18'000 unique combinations. For each pairing, optimal system sizes are computed using the single-roof simulator proposed in [4, 5]. It computes a robust sizing using a multivariate Chebyshev bound. Combinations resulting in PV sizes exceeding 10 kW or battery capacities above 20 kWh are excluded, as such systems are generally impractical for residential settings due to spatial and financial constraints. By splitting the data into train and test sets before combining the load and solar profiles, we ensure that only profiles which have never been seen before are contained in the test dataset. This approach results in a training dataset containing just above 10'000 samples.

To predict the optimal system sizing, we employ an MLP composed of four fully connected layers. The model is trained using the mean squared error loss function and the Adam optimizer with a learning rate of 0.001. The MLP is a suitable choice as it can learn robust mappings even for small datasets where more complex architectures are prone to overfitting.

4 Preliminary Results

The performance of the model is summarized in Table 1, which reports the mean squared error (MSE) and mean absolute error (MAE) for both battery and PV predictions on the test set. While the prediction errors are higher than those obtained through detailed simulations, the inference time of the neural network is reduced to mere milliseconds, making it suitable for fast decision-making applications. In comparison, the simulator proposed in [4] and used for labeling requires on average 40.58 seconds measured over 100 runs on a MacBook Pro with an Apple M1 chip. As a baseline, we use a Gradient Boosted Tree (XGBoost) trained on the same input features. While inference time remained fast, accuracy was lower.

In real-world deployment, PV panels are typically sold in discrete sizes, commonly between 350 W and 450 W per panel [3]. Consequently, a solar sizing error within this range corresponds to less than one panel. Battery systems are also purchased in coarse capacity increments. GivEnergy, a UK-based manufacturer, for instance offers models with 2.6 kWh, 5.2 kWh, and 9.5 kWh capacities [2]. Given this context, the model's predictions are sufficiently accurate for practical use, especially in early-stage system planning or as part of scalable tools for mass adoption.

Table 1: Performance metrics for battery and solar sizing

Metric	Battery (kWh)	Solar (kW)
XGBoost MSE	1.057	0.630
XGBoost MAE	0.734	0.601
MLP Test MSE	0.664	0.241
MLP Test MAE	0.614	0.364
MLP Inference Time (ms)	1.08 ($\pm$ 0.077)	
Simulation Duration (s)	40.58 ($\pm$ 1.18)	

5 Discussion

One of the key challenges is the limited availability of large load profile datasets due to privacy concerns. Although some datasets, such as the Pecan Street dataset [6], are available, they typically contain only tens to a few hundred samples, which is insufficient for robust model training. As a result, we rely on synthetic data to address this gap. Additionally, the optimal sizing of systems is highly dependent on weather, which is subject to significant variability. Given the rapid changes in weather patterns due to climate change, it remains uncertain whether historical data will continue to be relevant for future forecasting.

6 Conclusion

We present a novel approach for computing optimal battery and PV system sizes for residential buildings. By training an MLP on a diverse set of synthetic and augmented load and solar generation profiles, our model achieves accuracy while significantly reducing computation time. Given the discrete nature of the system components and the need for scalable planning tools, this approach offers a practical solution to quickly obtain sizing estimates.

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References

- [1] Center for Net Zero. 2024. Faraday: GenAI Modelling of Household Electricity Demand. <https://www.centreformetazero.org/technologies/faraday/>. Accessed: 2025-04-06.
- [2] GivEnergy. 2024. Technical Specifications. <https://givenenergy.co.uk/products/battery-storage/>. Accessed: 2025-04-06.
- [3] Hive. 2024. Solar Panel Sizes in the UK. <https://www.hivehome.com/discover-hive/smart-energy/solar-panel-sizes-in-the-uk>. Accessed: 2025-04-06.
- [4] Brad Huang. 2023. Robust Sizing. <https://github.com/BradHuang1999/Robust-Sizing>. Accessed: 2025-04-06.
- [5] Brad Guanqiao Huang, Fiodar Kazhamiaka, and Srinivasan Keshav. 2021. Sizing solar panels and storage for multiple roofs. In *Proceedings of the Twelfth ACM International Conference on Future Energy Systems*. 84–94.
- [6] Pecan Street Inc. 2024. Dataport. <https://www.pecanstreet.org/dataport/>. Accessed: 2025-04-06.
- [7] Najwa Syahirah Mohamed Nor Izam, Zarina Itam, Wong Leong Sing, and Agusril Syamsir. 2022. Sustainable development perspectives of solar energy technologies with focus on solar Photovoltaic—A review. *Energies* 15, 8 (2022), 2790.
- [8] Fiodar Kazhamiaka, Yashar Ghiassi-Farrokhi, Srinivasan Keshav, and Catherine Rosenberg. 2018. Robust and Practical Approaches for Solar PV and Storage Sizing. In *Proceedings of the Ninth International Conference on Future Energy Systems (Karlsruhe, Germany) (e-Energy '18)*. Association for Computing Machinery, New York, NY, USA, 146–156. doi:10.1145/3208903.3208935
- [9] PVWatts. 2024. A Performance Calculator for Grid-Connected PV Systems. <https://pvwatts.nrel.gov/>. Accessed: 2025-04-06.